

IP Addressing & Subnetting

Answers

Q1. IP Address Classes

List the IP address classes (A-E) with one real-world example each, in a table.

Class	First Octet Range	Default Mask	Purpose	Real-World Example
A	1 - 126	255.0.0.0 (/8)	Very large networks (about 16 million hosts each)	Large ISPs / global corporations, e.g. a telecom backbone network
B	128 - 191	255.255.0.0 (/16)	Medium to large networks (about 65,534 hosts)	University or large college campus network
C	192 - 223	255.255.255.0 (/24)	Small networks (254 hosts)	Home WiFi router, small office LAN
D	224 - 239	N/A (multicast)	Multicast: one sender to many receivers	IPTV / live video streaming; routing protocols like OSPF (224.0.0.5)
E	240 - 255	N/A (reserved)	Reserved for experimental / future use	Not used on real devices; used only for research and testing

Note: 127.x.x.x is reserved for loopback (localhost), so Class A usable range is 1-126.

Q2. Network, Broadcast and Host Range

Given IP 192.168.10.45 with mask 255.255.255.0, find network address, broadcast address and valid host range.

Step 1: Convert to binary

	Octet 1	Octet 2	Octet 3	Octet 4
IP address	11000000	10101000	00001010	00101101
Subnet mask	11111111	11111111	11111111	00000000
AND result (Network)	11000000	10101000	00001010	00000000

Step 2: Network address = IP AND Mask = 192.168.10.0

Step 3: Broadcast address: mask is /24, so 24 bits are network and 8 bits are host. Set all host bits to 1: 11111111 = 255, giving 192.168.10.255

Step 4: Valid host range: first host = network + 1 = 192.168.10.1; last host = broadcast - 1 = 192.168.10.254

Step 5: Total hosts = $2^8 - 2 = 254$

Network: 192.168.10.0 | Broadcast: 192.168.10.255 | Hosts: 192.168.10.1 to 192.168.10.254

Q3. Subnetting Plan for a Food Delivery Startup (10.0.0.0/24)

Create 4 subnets: Delivery Partners, Restaurants, Customers, Admins.

Step 1: Subnets needed = 4. Bits to borrow: $2^2 = 4$, so borrow 2 bits.

Step 2: New prefix = /24 + 2 = /26, mask = **255.255.255.192**.

Step 3: Block size = 256 - 192 = 64 addresses per subnet; usable hosts = 64 - 2 = 62.

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 - 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 - 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 - 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 - 10.0.0.254	10.0.0.255

Each group gets a unique /26 subnet with 62 usable hosts.

Q4. College WiFi Network (172.16.0.0/16)

At least 20 subnets are needed. Find the new subnet mask and hosts per subnet.

Step 1: Bits to borrow. Find n such that 2^n is at least 20. $2^4 = 16$ (too small); $2^5 = 32$ (enough). So $n = 5$.

Step 2: New prefix. $/16 + 5 = /21$.

Step 3: New mask. 21 network bits = 11111111.11111111.11111000.00000000 = **255.255.248.0**.

Step 4: Hosts per subnet. Host bits = $32 - 21 = 11$. Hosts = $2^{11} - 2 = 2048 - 2 = \mathbf{2046}$.

Subnets created = 32 (block size 8 in the third octet: 172.16.0.0, 172.16.8.0, 172.16.16.0, ...)

New mask: 255.255.248.0 (/21) | Hosts per subnet: 2046

Q5. Why Does 192.168.1.300 Not Work?

A friend's device with IP 192.168.1.300 cannot connect. Identify the error.

Error: The IP address is invalid. An IPv4 address has four octets, and each octet is 8 bits, so it can only be between 0 and 255. The last octet here is 300, which is out of range, so the device cannot use this address and fails to connect.

Valid format: In the 192.168.1.0/24 network (mask 255.255.255.0), valid host addresses are 192.168.1.1 to 192.168.1.254. Examples of valid addresses: **192.168.1.30**, 192.168.1.100, 192.168.1.200. (192.168.1.0 is the network address and 192.168.1.255 is the broadcast address, so they should not be assigned to a device.)

Fix: Change the last octet to a number from 1 to 254 (probably a typo, e.g. 192.168.1.30), or set the device to obtain an IP automatically via DHCP.